Stage 3 Outer Race Fault — Gearbox Bottleneck, Part Shortage, Monitoring Active

Asset: AST_GBX_001 (Gearbox) | Bearing: BRG_011 (SKF22318)
Fault Mode: outer_race_fault | ISO Stage: Stage 3
Learned Case — Knowledge Agent Retrieval Document | DRO Baseline
CASE_004 | Outcome: PENDING | Opened: 2026-05-15 | Closed: In Progress

1. Detection Summary

Detection Date	2026-05-15
Detected By	monitoring_agent
Anomaly Score	0.89 (high confidence)
vib_rms_mm_s	7.2 mm/s (baseline mean: 2.8 mm/s — z-score: 11σ)
kurtosis	7.0 (threshold: 5.0 — exceeded, fault confirmed)
temp_c	86°C (baseline mean: 62°C — rise: 24°C)
bpfo_energy	3.5× (stage_3 boundary exactly — Stage 3 classified)
signal_quality_score	0.96–0.97 (clean signal)
Asset Context	criticality=high, is_bottleneck=true, downtime_cost=INR 18,000/hr, iso_zone=D
Financial Exposure	0.85 × 8h × INR 18,000/hr = INR 122,400 if run to failure

2. Agent Diagnosis

Primary Diagnosis	outer_race_fault — Stage 3
Confidence	0.85
Reasoning	kurtosis 7.0 > FT_001 threshold 5.0. bpfo_energy 3.5× at stage_3 boundary. Temperature rise 24°C consistent. Bottleneck asset — risk escalated.
RUL Estimate	5–9 days from Stage 3 detection (based on CASE_001 and other KB cases)
Risk Level	CRITICAL — bottleneck asset, high downtime cost, Zone D

3. Action Taken

Inventory Check	PART_003 (SKF22318-E) qty_on_hand = 0. Reserved = 0.
Lead Time Standard	30 days (exceeds RUL — cannot wait for standard delivery)
Lead Time Expedited	12 days (exceeds RUL lower bound of 5 days — risk window)
Recommendation Changed To	Emergency procurement + derate + 2-hour monitoring interval
Interim Actions	Load reduced 15–20%. Monitoring at 2h intervals. Maintenance window pre-booked Day 13.
Work Order	WO_004 — priority CRITICAL
LOTO	LOTO_004 — all shift supervisors briefed on Zone D (no approach without LOTO)

4. Findings at Inspection / Teardown

Gearbox not yet opened — awaiting PART_003 delivery (expected 2026-05-27). No teardown findings available. Bearing is being monitored every 2 hours. Load derating of approximately 15% is in effect operationally. Procurement order placed with SKF India, expedited delivery. Repair window pre-booked for 2026-05-28.

5. Outcome

Current Status	IN PROGRESS — monitoring active
Post-Repair Measurements	Not yet available
QA Result	Pending
Actual Duration	Pending
Actual Cost	INR 12,000 committed (estimated, excluding part cost)
Expected Part Arrival	2026-05-27 (expedited)
Expected Repair Date	2026-05-28

6. Root Cause Confirmed

Root Cause	Pending teardown — outer race fault confirmed by telemetry
Preliminary Contributing Factors	Rapid escalation pattern suggests possible contamination or fatigue loading. Oil last changed 2026-02-01 — 108 days. To be confirmed at teardown.

7. Lessons Learned

- When part lead time exceeds RUL lower bound, recommendation must change from 'schedule repair' to 'expedite procurement + monitor + derate'. Prescriptive Agent handled this correctly.
- SKF22318-E is a long-lead item (30 days standard) on a critical bottleneck asset. Safety stock of at least 1 unit should be maintained. Zero safety stock on critical spare is unacceptable risk.
- Zone D classification: no approach for any reason without LOTO_004 fully applied. All shift supervisors must be briefed at handover.
- Financial exposure at INR 122,400 potential loss justified emergency procurement premium over waiting for standard delivery.
- Derate + increased monitoring is the correct holding strategy when part is unavailable — do not run full load on a Stage 3 bearing.